Closure Dynamics: A Variational and Gradient Formulation of Constraint-Manifold Evolution

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Abstract

Papers V–XI developed a structural constraint framework for particle mass, gauge structure, confinement, continuous geometry, and a gravitational analogy, but did not include a time-evolution law. The present paper introduces minimal dynamics into the closure framework by defining state evolution as gradient flow of the closure functional $\mathcal{F}$.

Under this evolution, the closure functional decreases monotonically along trajectories, closure-stable states ($\mathcal{F} = 0$) are fixed points, and distinct attractor basins correspond to distinct particle classes. Disconnected configurations are dynamically unstable: they flow toward connected composites, providing a dynamical interpretation of confinement. A conservative variational extension (Lagrangian formulation) is introduced as a second-order generalisation.

The evolution parameter t is not identified with physical time. No relativistic invariance, quantum field theory, scattering amplitudes, or coupling constants are derived. The paper establishes the minimal dynamical layer consistent with the closure-constraint structure, providing the formal transition from a purely structural framework to one that admits evolution.

1 Introduction

The preceding papers in this programme [Smart, 2026b,c,d,a,e,f] developed a constraint-based framework in which physical states correspond to the closed state set $\mathcal{S}_{\mathrm{cl}}$, mass is the closure residual, and the constraint manifold $\mathcal{M}_{\mathrm{cl}}$ carries geometric structure. Throughout that development, the framework was purely structural: it specified *which* states are admissible and *what* properties they carry, but not *how* states evolve.

The present paper fills this gap by introducing a minimal evolution law. The dynamics is defined as gradient flow of the closure functional: states evolve in the direction that most rapidly reduces constraint violation. This is the simplest dynamical extension

consistent with the closure structure, and it produces the expected behaviour: closure-stable states are attractors, disconnected configurations are unstable, and the constraint manifold governs the geometry of admissible motion.

Scope and epistemic status

This paper introduces *minimal dynamics*: a gradient-flow evolution law and its variational extension. It does not introduce:

- physical time (the evolution parameter t is auxiliary),
- relativistic invariance or Lorentz covariance,
- quantum field theory or operator-valued dynamics,
- scattering amplitudes, decay rates, or coupling constants.

The dynamics is presented as the simplest evolution law compatible with the closure structure. Whether it can be extended to a full physical dynamics is an open question.

2 Time-Dependent State Space

We extend the admissible state space to include a time parameter:

$$\psi = \psi(t), \qquad t \in \mathbb{R} \tag{1}$$

where $\psi(t) \in \mathcal{S}_{\text{adm}}$ for all t. The state traces a curve in the admissible domain as the evolution parameter advances.

Remark 1 (Status of t)**.** *The parameter t is introduced as an auxiliary evolution parameter governing relaxation within the constraint framework. It is not identified with physical coordinate time, proper time, or any relativistic time parameter. The identification of t with a physical quantity would require additional structure (e.g. a metric signature or Lorentz group action) not present in this paper.*

3 Gradient Flow Dynamics

3.1 The evolution law

We define the evolution of states as gradient flow of the closure functional $\mathcal{F}$:

$$\frac{d\psi}{dt} = -\nabla\mathcal{F}(\psi) \tag{2}$$

where $\nabla\mathcal{F}$ is the gradient of $\mathcal{F}$ with respect to the inner product on the state space V.

3.2 Constrained evolution

Since $\psi(t)$ must remain in $\mathcal{S}_{\text{adm}}$ (e.g. normalised), the flow is projected onto the admissible domain:

$$\frac{d\psi}{dt} = -\Pi_{\mathcal{S}_{\text{adm}}}(\nabla\mathcal{F}(\psi)) \tag{3}$$

where $\Pi_{\mathcal{S}_{\text{adm}}}$ denotes the orthogonal projection onto the tangent space of $\mathcal{S}_{\text{adm}}$ at ψ with respect to the ambient inner product.

3.3 Monotonic decrease

Proposition 2 (Monotonic descent)**.** *Along trajectories of the projected gradient flow* (3)*, the closure functional decreases monotonically:*

$$\frac{d\mathcal{F}}{dt} = \langle \nabla\mathcal{F}, \dot{\psi} \rangle = -\|\Pi_{\mathcal{S}_{\text{adm}}}(\nabla\mathcal{F})\|^2 \leq 0 \tag{4}$$

Equality holds if and only if $\Pi_{\mathcal{S}_{\text{adm}}}(\nabla\mathcal{F}(\psi)) = 0$*, i.e. at a constrained critical point.*

Proof. By the chain rule, $d\mathcal{F}/dt = \langle \nabla\mathcal{F}, d\psi/dt \rangle$. Substituting (3): $d\mathcal{F}/dt = -\langle \nabla\mathcal{F}, \Pi_{\mathcal{S}_{\text{adm}}}(\nabla\mathcal{F}) \rangle$. Since $\Pi_{\mathcal{S}_{\text{adm}}}$ is an orthogonal projection, $\langle \nabla\mathcal{F}, \Pi_{\mathcal{S}_{\text{adm}}}(\nabla\mathcal{F}) \rangle = \|\Pi_{\mathcal{S}_{\text{adm}}}(\nabla\mathcal{F})\|^2 \geq 0$. □

3.4 Interpretation

The gradient flow drives the system toward lower values of $\mathcal{F}$:

- $\mathcal{F}(\psi)$ decreases monotonically along trajectories.
- States with $\mathcal{F}(\psi) = 0$ — i.e. states in $\mathcal{S}_{\text{cl}}$ — are fixed points.
- The flow represents constraint relaxation: the system evolves in the direction of steepest descent toward closure satisfaction.

This is the simplest dynamics consistent with the closure structure: it uses only $\mathcal{F}$ and the inner product, introduces no additional parameters, and automatically identifies $\mathcal{S}_{\text{cl}}$ as the set of attractors. The gradient flow should be interpreted as a coarse-grained or relaxation dynamics, not a fundamental microscopic evolution law.

4 Fixed Points and Stability

Definition 3 (Dynamical fixed point)**.** *A state* $\psi^* \in \mathcal{S}_{\text{adm}}$ *is a fixed point of the gradient-flow dynamics if:*

$$\Pi_{\mathcal{S}_{\text{adm}}}(\nabla\mathcal{F}(\psi^*)) = 0 \tag{5}$$

All states in $\mathcal{S}_{\text{cl}}$ satisfy $\mathcal{F}(\psi^*) = 0$ and are therefore fixed points of the constrained flow: the projected gradient vanishes at a constrained minimum of $\mathcal{F}$ on $\mathcal{S}_{\text{adm}}$.

4.1 Local stability

The stability of a fixed point is determined by the constrained Hessian:

$$H(\psi) = \Pi_{\mathcal{S}_{\text{adm}}}\big(\nabla^2\mathcal{F}(\psi)\big)\,\big|_{T_\psi\mathcal{S}_{\text{adm}}} \tag{6}$$

restricted to the tangent space of $\mathcal{S}_{\text{adm}}$ at ψ.

Remark 4 (Stability criterion)**.** *A fixed point* ψ^* *is a stable attractor if* $H(\psi^*)$ *is positive semi-definite on* $T_{\psi^*}\mathcal{S}_{\text{adm}}$*: all nearby perturbations within the admissible domain are driven back toward* ψ^* *by the flow. Saddle points of* $\mathcal{F}$ *(with indefinite Hessian) are unstable fixed points.*

5 Attractor Structure

Definition 5 (Attractor basin)**.** *The attractor basin of a stable fixed point $\psi^* \in \mathcal{S}_{\mathrm{cl}}$ is:*

$$B(\psi^*) = \left\{ \psi_0 \in \mathcal{S}_{\mathrm{adm}} \;\middle|\; \lim_{t\to\infty} \psi(t;\psi_0) = \psi^* \right\} \tag{7}$$

where $\psi(t;\psi_0)$ is the trajectory of the gradient flow starting from ψ_0.

Within the framework, distinct attractor basins provide the dynamical analogue of distinct particle classes: initial configurations in basin $B(\psi^*)$ relax toward the closure-stable state ψ^*. The particle type is determined by which attractor the initial configuration flows toward.

Remark 6. *The attractor-basin structure depends on the global geometry of $\mathcal{F}$ on $\mathcal{S}_{\mathrm{adm}}$. In general, the basin boundaries are the stable manifolds of saddle points. A full characterisation of the basin structure for the 600-cell closure functional is beyond the scope of the present paper.*

6 Mass and Dynamics

Mass is defined as the closure residual (Papers V, VII):

$$m = \kappa\, d(\psi, \mathcal{M}_{\mathrm{cl}}) \tag{8}$$

Under gradient flow, this acquires a dynamical interpretation:

- **Relaxation toward closure:** states with $\mathcal{F} > 0$ evolve toward $\mathcal{M}_{\mathrm{cl}}$. The closure residual $\Delta(\psi) = d(\psi, \mathcal{M}_{\mathrm{cl}})$ decreases along trajectories (since $\mathcal{F}$ decreases and Δ is related to $\mathcal{F}$). The relation between Δ and $\mathcal{F}$ is structural rather than theorem-level: both decrease under relaxation toward closure, but no exact functional identity between them is assumed in this paper.
- **Mass and the closure landscape:** mass is associated with the local structure of the closure landscape around $\mathcal{M}_{\mathrm{cl}}$, rather than requiring states to lie permanently off the manifold in equilibrium. Within the present framework, particle states are associated with closure-stable configurations of the dynamical system; the precise physical interpretation of these configurations beyond the minimal dynamical layer remains open.
- **Motion on $\mathcal{M}_{\mathrm{cl}}$:** once a trajectory reaches $\mathcal{M}_{\mathrm{cl}}$ (where $\mathcal{F} = 0$), the gradient vanishes and the state is stationary under the dissipative gradient flow. Motion along $\mathcal{M}_{\mathrm{cl}}$ requires the conservative (Lagrangian) extension of Section 8.

7 Confinement in Dynamics

Disconnected configurations (Paper VIII) have $\mathcal{F} > 0$ due to the gap penalty $\Xi > 0$. Under gradient flow:

- disconnected states are *not* fixed points (they have $\nabla\mathcal{F} \neq 0$),

- they flow toward configurations with lower $\mathcal{F}$,
- the flow drives them toward connected composites (where $\Xi = 0$),
- they cannot stabilise independently.

This provides a *dynamical interpretation of confinement*: disconnected quark-like states are unstable under closure dynamics. They are not held together by a force; they are driven toward connected composites by the gradient of the constraint functional.

Remark 7. *This dynamical confinement is a consequence of the gradient-flow structure and the gap penalty in $\mathcal{F}$. It does not invoke a non-Abelian gauge field, a confining potential, or colour flux tubes. It is a structural result within the closure framework.*

8 Variational / Lagrangian Extension

The gradient flow (2) is a first-order, dissipative dynamics: it drives the system toward fixed points but does not describe oscillatory or conservative motion. To extend the framework to conservative dynamics, we introduce a Lagrangian formulation.

8.1 The closure Lagrangian

Definition 8 (Closure Lagrangian)**.** *The closure Lagrangian is:*

$$\mathcal{L}(\psi, \dot{\psi}) = \frac{1}{2}\|\dot{\psi}\|^2 - \mathcal{F}(\psi) \tag{9}$$

where $\|\dot{\psi}\|^2 = \langle \dot{\psi}, \dot{\psi} \rangle$ is the kinetic term and $\mathcal{F}(\psi)$ plays the role of the potential. This choice corresponds to the simplest quadratic kinetic term compatible with the ambient inner-product structure; it is not claimed to be unique.

8.2 The action and equations of motion

The action is:

$$S[\psi] = \int_{t_1}^{t_2} \mathcal{L}(\psi, \dot{\psi})\, dt \tag{10}$$

Stationarity of the action ($\delta S = 0$) yields the Euler–Lagrange equation:

$$\ddot{\psi} = -\nabla \mathcal{F}(\psi) \tag{11}$$

This is a second-order dynamical system in which $\mathcal{F}$ acts as a potential energy landscape. States oscillate around minima of $\mathcal{F}$ (closure-stable configurations) rather than dissipating toward them.

8.3 Constrained Lagrangian dynamics

For evolution constrained to $\mathcal{S}_{\text{adm}}$, the equations of motion become:

$$\Pi_{\mathcal{S}_{\text{adm}}}\left(\ddot{\psi}\right) = -\Pi_{\mathcal{S}_{\text{adm}}}(\nabla \mathcal{F}(\psi)) \tag{12}$$

with the constraint that $\psi(t) \in \mathcal{S}_{\text{adm}}$ enforced throughout.

8.4 Interpretation

- **Gradient flow** (first-order): dissipative relaxation toward closure. Models constraint satisfaction.
- **Lagrangian dynamics** (second-order): conservative motion on the $\mathcal{F}$ landscape. Models evolution within and around closure-stable configurations.

Both formulations share the same fixed points ($\mathcal{S}_{\text{cl}}$) and the same stability structure (determined by $H(\psi)$). They differ in their dynamical character: the gradient flow is irreversible; the Lagrangian dynamics is time-reversible.

Remark 9. *The Lagrangian (9) is the simplest conservative extension of the closure structure. It uses only $\mathcal{F}$ and the inner product, introduces no additional parameters, and reduces to the gradient flow in the overdamped limit ($\ddot{\psi} \to 0$). Whether a more physical Lagrangian (incorporating, e.g., gauge coupling or relativistic structure) can be constructed from the closure framework is an open question.*

9 Two Modes of Motion

The dynamics defined in this paper produces two distinct modes of motion:

1. **Transverse motion** (toward/away from $\mathcal{M}_{\text{cl}}$): evolution in the direction normal to the constraint manifold. Under gradient flow, this is relaxation toward closure. Under the Lagrangian, it is oscillation around $\mathcal{M}_{\text{cl}}$. This mode is associated with mass ($\Delta > 0$) and closure residual.
2. **Tangential motion** (along $\mathcal{M}_{\text{cl}}$): evolution within the constraint manifold. Under the Lagrangian, states on $\mathcal{M}_{\text{cl}}$ can move along geodesic-like paths (Paper X [Smart, 2026e]). Under gradient flow, tangential motion does not occur (the gradient vanishes on $\mathcal{M}_{\text{cl}}$).

The separation into transverse and tangential modes provides a dynamical counterpart to the structural distinction between mass (distance from $\mathcal{M}_{\text{cl}}$) and geometry (motion on $\mathcal{M}_{\text{cl}}$).

Remark 10 (Two levels of dynamics)**.** *The gradient-flow dynamics should be interpreted as a relaxation process toward closure-stable configurations. It is not claimed to represent fundamental microscopic dynamics. A fully physical evolution law would require additional structure — potentially building on the variational formulation, incorporating gauge coupling or relativistic invariance — and represents the primary open problem for the programme's dynamical layer.*

10 Limitations

- **The evolution parameter t is not physical time.** It is an auxiliary parameter. Identification with physical time requires additional structure (metric signature, Lorentz covariance) not present here.

- **No relativistic invariance.** The dynamics is defined on a state space with Euclidean inner product, not on a Lorentzian manifold.
- **No quantum dynamics.** The evolution is classical (deterministic gradient flow or Lagrangian mechanics). Quantisation is not addressed.
- **No scattering or decay.** The dynamics describes relaxation and oscillation around attractors, not scattering processes between states.
- **No coupling constants derived.** The mass parameter κ and the closure-functional weights remain structural parameters.
- **The Lagrangian is minimal.** It is the simplest conservative extension; more physical Lagrangians may be needed.
- **Basin structure not fully characterised.** The global attractor-basin geometry depends on $\mathcal{F}$ and is not computed here.

11 Conclusion

The closure framework, previously purely structural, now admits a minimal dynamical extension. States evolve via gradient flow of the closure functional: $\mathcal{F}$ decreases monotonically, closure-stable states are attractors, and disconnected configurations are dynamically unstable (confinement). A conservative Lagrangian extension provides second-order dynamics with oscillatory motion around closure-stable configurations.

The dynamics separates into two modes: transverse (toward/away from $\mathcal{M}_{\mathrm{cl}}$, associated with mass) and tangential (along $\mathcal{M}_{\mathrm{cl}}$, associated with geometry). This decomposition connects the dynamical evolution to the structural properties established in Papers V–X.

The evolution parameter t is not identified with physical time, and no relativistic or quantum structure is introduced. In this sense, the present paper marks the transition from a purely structural closure framework to a minimally dynamical one: not yet a full physical theory, but no longer a static correspondence framework alone.

References

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